CHAPTER

22

Microleakage

The goal of operative dentistry, undoubtedly, is to restore the tooth to its form and functions. One of the requisites of a restorative material is to adapt itself to cavity walls. Among the various restorative materials currently used, and despite of the tremendous improvements in means and technologies, none of the material could actually bond chemically with the tooth surface. The gap left between the cavity walls and the restorative material plays an important role in the prognosis of the restorative treatments. In the past, pulpal reactions to dental procedures were thought to be induced by mechanical irritation, like heat, vibration, galvanism, etc. and/or chemical irritation by the restorative material and its components. Various authors have demonstrated that the bacterial leakage was a greater threat to the pulp than the toxicity of restorative materials. Since then the concept of microleakage has drawn widespread attention. Different authors have termed 'leakage' as marginal percolation, liquid diffusion, fluid exchange, capillary penetration, etc.

Microleakage can be defined as "*the clinically undetectable passage of bacteria and bacterial products, fluids, molecules or ions from the oral environment along the various gaps present in the cavity–restoration interface*".

This interface can be composed of tooth, smear layer, filling and the cementing medium (if any). There are three possible routes of microleakage: (1) Within or via the smear layer, (2) between the smear layer and the cavity varnish/cement, and (3) between the cavity varnish/cement and the restoration (Fig. 22.1). The cavity–restoration interface is not a fixed, inert or an impenetrable border; but a dynamic microcrevice, which allows free movement of ions and molecules. It has been established that a minimum of 10.0 μm space is left at the tooth–restoration interface even after employing the adhesive liners and materials.

Microleakage is hence a serious clinical problem that requires thorough analysis and discussion because all restorative materials exhibit varying degrees of leakage at the margins.

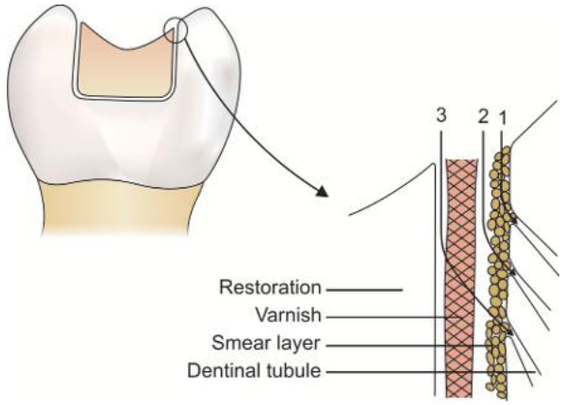


Fig. 22.1: Diagram showing cavity–restoration interface and the possible routes of microleakage: (1) Within or via the smear layer, (2) through the gap between the varnish and smear layer, and (3) between the varnish and restoration

CLINICAL IMPLICATIONS OF MICROLEAKAGE

1. ***Postoperative sensitivity:*** Microleakage often results in dentin sensitivity by creating a direct communication between oral fluids and the pulp. Change in local ionic concentrations, caused by leakage of acids and other substances, facilitates movement of fluid in the dentinal tubules, leading to pain. Postoperative sensitivity is more pronounced in proximal/cervical cavities and more common with resin restorations.
2. ***Secondary caries/recurrent caries:*** Carious lesions that exist along the walls of a restored cavity can be attributed to the existing microleakage at the tooth–restoration interface. Bacteria with an average diameter less than 1.0 μm can easily penetrate such gaps. A usual gap of more than 10 μm provides enough space to house nutrients for thriving of bacteria; hence, inviting recurrent caries.
3. ***Pulpal pathology:*** Marginal gaps around restorations may become sufficiently wide to allow in-growth of bacteria. These bacteria may produce a number of inflammatory components, which

penetrate the dentinal tubules, affecting the underlying pulp; subsequently, causing pulpal diseases.

4. ***Marginal discoloration:*** Marginal discoloration, commonly encountered in esthetic restorative materials, may lead to accumulation of subsurface interfacial staining. It is difficult to eliminate such staining by conventional means. A marked boundary is created around the restoration, which necessitates its replacement.
5. ***Breakdown and dissolution*** of certain materials, like cement lutes.

RESTORATIVE MATERIALS AND MICROLEAKAGE

Microleakage is a useful indicator of the microscopic and anatomic aspects of bond between restorative material and the tooth. Black has stressed that the filling materials should be closely adapted to the cavity walls and should withstand thermal changes once placed in the oral environment. However, no restorative material can chemically bond to the tooth structure providing perfect seal, which is capable of withstanding the temperature fluctuations of oral environment. Good marginal adaptation requires adequate physical and mechanical properties of the filling material and its careful manipulation. The properties of restorative materials that contribute largely to microleakage are:

i. Coefficient of thermal expansion
ii. Polymerization shrinkage
iii. Adhesion

Other properties like creep, elasticity, resistance to fatigue failures, solubility, etc. also contribute to a minor extent.

i. ***Coefficient of thermal expansion (CTE):*** It is the change in length per unit length of a material per degree change in temperature. Both linear and volumetric coefficient of thermal expansion are important in restorative materials. On cooling (reduction in temperature), there occurs contraction of the material; conversely, heating (increasing temperature) leads to expansion. Ideally, a restorative material should have a CTE, which closely matches that of tooth structure, so that there occurs an equal expansion and contraction of both. In case of varying CTE between the two, one may expand/contract more than the other during a change in temperature, leading to a leaky restoration margin, i.e. microleakage. The greater the difference between CTE of two materials, greater will be the microleakage. From the Table 22.1, one can evaluate that an acrylic restoration will show a greater microleakage as compared to glass-ionomer or direct gold restorations. It has also been emphasized that marginal percolation can be because of the changes in volume of fluid occupying the space between tooth and the restoration, subsequent to thermal changes.

Table 22.1: Linear coefficient of thermal expansion of dental materials relative to tooth enamel

Material	*CTE ($\times 10^{-6}$/°C)*
Tooth (crown)	11.4
Dentin	8.3
Aluminous porcelain	6.6
Pure titanium	8.5
Type II glass-ionomer	11.0
Pure gold	14.0
Silver amalgam	25.0
Composites	20.0–25.0
Denture resin	81.0
Inlay wax	400.0

ii. ***Polymerization shrinkage or volumetric shrinkage:*** It occurs with polymeric materials. When the monomer chains are polymerized to form polymer chain, there occurs a decrease in volume and an increase in density. (For the same mass, if density increases, volume has to reduce since Density = Mass/Volume). The shrinkage tends to pull the material away from the cavity walls. If an intermediate adhesive resin is used to bond the restorative resin to the tooth and the contraction stresses generated are very high, there may occur a break in the adhesive bond resulting in promotion of microleakage. The polymerization shrinkage of different polymeric materials is depicted in Table 22.2.

iii. ***Adhesion:*** It is the attraction of the molecules of two different substances to each other, when they are brought in close contact. Lack of adhesion between restorative material and the tooth may allow penetration of the external materials at a microscopic level. Adhesion is influenced by the wetting capabilities, surface energy, presence of water and smear layer, difference in composition

Table 22.2: Polymerization shrinkage of different resins

Resin	*Polymerization shrinkage*
Acrylic heat cure	7%
Acrylic cold cure	3.5%
Conventional composite	1.5–2.0%
Organic composite	2.5–3.5%
Microfilled composite	1.3–1.5%
Hybrid composite	2.2–2.5%

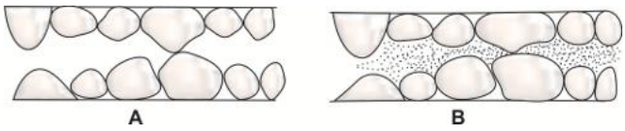


Fig. 22.2: **(A)** Inadequate contact because of suface irregularities could lead to microleakage; **(B)** A low viscosity material fills the gap and reduces microleakage

of enamel and dentin, surface roughness, etc. Adhering of two solid surfaces is very difficult. This can be overcome to a small extent by using an intermediate layer of fluid or a low viscosity material to fill the irregularities and the space existing between the two (Fig. 22.2A, B).

The leakage pattern at the tooth–restoration interface is also influenced by the manipulation of the material. Poor packing, condensation and insertion of restorative materials and/or poor cavity design may lead to increased dimensional changes, early dissolution of the materials and poor marginal fit. The restorative materials require the operator's skill in adapting the material to the cavity walls, retention areas and the cavosurface margins. Various situations in which microleakage is promoted because of the negligence of the operator can be cited in our day-to-day practice, e.g. leaving the margins of a cast inlay unburnished, exposing a thin cement line to the oral environment, etc. Failure to isolate the cavity during insertion of composite resin may also facilitate leakage. Surface contamination may result in an inadequate bond between resin and the tooth, resulting in microleakage.

Role of Smear Layer

During cavity preparation, the microcrystalline debris may get embedded within the denatured collagen of dentin; these deposits formed on the cut surfaces is known as '*smear layer*'. It is generally 1.0–2.0 µm thick and may consist of blood, saliva, bacteria, enamel and dentin particles. Some of the initial cutting debris may also be pushed into the dentinal tubules by almost 1.0–5.0 µm forming the '*smear plugs*'. The smear layer and the smear plugs serve as a functional unit (Figs 22.3 and 22.4) to reduce dentin permeability and thereby protect the pulp. However, the role of smear layer in preventing or promoting microleakage is still unclear. There is divergence of opinion as regard the role of smear layer in microleakage.

One opinion is to leave the smear layer intact on the tooth surface, because it may act as a physical barrier in preventing the ingress of bacteria and their products into the underlying dentin. Pashley (1991) has reported a decrease of 86% in the permeability of dentin, when the smear layer is present. Clinicians

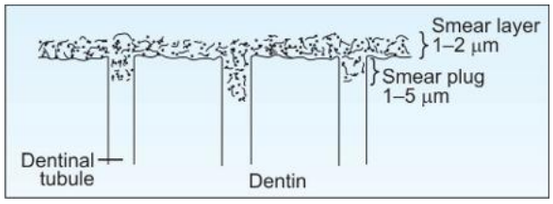


Fig. 22.3: Smear layer and plugs

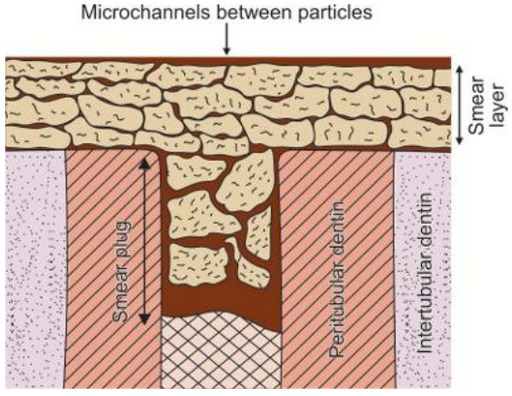


Fig. 22.4: Enlarged view of smear layer and plug around peritubular and intertubular dentin

supplementing this view should keep in mind that the smear layer is acid labile at a pH of 6.0–6.8 and less. It is inadvertently destroyed when the pH drops, e.g. around plaque accumulated areas, or by proteolytic enzymes excreted by certain micro-organisms or by acid etching. Smear layer on an exposed root surface dissolves more readily as compared to one present on the occlusal surface.

A few authors opine removal of the smear layer because of the belief that the smear layer may contain many bacteria. It has been demonstrated that the smear layer may be capable of preventing diffusion of bacteria, but not the bacterial products. A decrease in microleakage was observed when the smear layer was absent (Removal of smear layer permits a good adaptation of the restorative material to the cavity walls).

It is preferred to remove the natural smear layer and replace it with a '*sterile, inert and non-toxic*' synthetic smear layer. Research in this area has shown promising results. The smear layer is partially removed with weak acid leaving smear plugs. Weak acids, such as, tannic acid and polyacrylic acid may remove only the smear layer, leaving behind intact smear plugs, thereby allowing better adaptation of the restorative material to the cavity walls. The newer generation dentin bonding systems removes only the smear layer while retaining intratubular plugs.

Factors Controlling Bacterial Penetration at Tooth–Restoration Gaps

Bacterial entry into the tooth–restoration gaps is affected by the size and nature of the gap, defence mechanisms of the host and also quality of the restorative material. A schematic representation of this relationship is shown in Fig. 22.5.

1. ***Size and nature of the gap:*** The size of the gap at the tooth–restoration interface varies, ranging from 10–50 µm. Even a small gap of 10 µm is sufficient for the entry of oral micro-organisms. These gaps are usually sufficiently wide to house bacteria, provide continuous nutrition and facilitate their growth. After establishing themselves in this area, the organisms may produce a number of components, which on penetration through the dentinal tubules could adversely affect the underlying dental pulp.
   Nature of the gaps cannot be described in physiochemical terms. However, the self-sealing capabilities of the restorative material to a certain extent can reduce bacterial penetration. Self-sealing may occur because of deposition of mineral salts of low solubility in the space, accumulation of corrosion products, calcification of the debris around the margins, etc.
2. ***Host defense factors:*** Natural defense mechanisms of the tooth, like sclerosis of dentinal tubules and reparative dentin formation, decrease bacterial penetration. The hydrostatic pressure of the pulp is higher as compared to the outside pressure of the oral cavity. This pressure difference pushes the dentinal fluid outwards, which opposes the bacteria and their products moving inwards, though only to a negligible extent. Some plasma proteins in the dentinal fluid may act as antibacterial; whereas, large molecular weight proteins, like fibrinogen, make dentin less permeable to bacteria. Presence of smear plugs also decreases the diffusion of bacteria to the pulp. Hence, the process of acid etching, which removes these plugs and opens dentinal tubules, increases their chances of penetration. Alteration of the chemical structure of dentin by leaching of tin/mercury ions from amalgam or fluoride ions from glass-ionomer also checks bacterial diffusion.
3. ***Quality of the restorative material:*** Certain restorations, like glass-ionomers, compomers, etc. may release fluorides into the gaps. Fluorides have an antimicrobial effect and thereby reduce ingress of bacteria into the gaps. The elements released from amalgam (corrosion products) significantly close the tooth–restoration gap. Resin cements used in indirect restorations effectively close the gap.

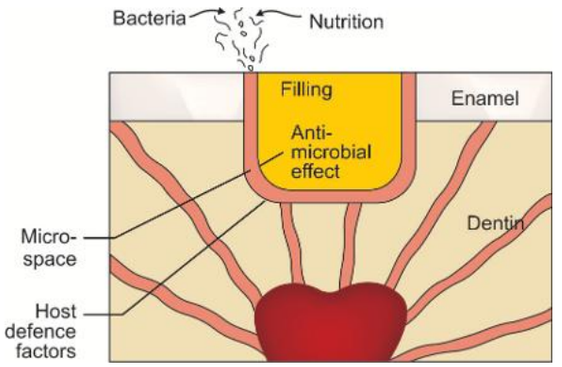


Fig. 22.5: Factors affecting bacterial penetration into marginal gaps

MICROLEAKAGE AND RESTORATIVE MATERIALS

a. Microleakage around Amalgam Restorations

Amalgam when freshly condensed does not adapt closely to the walls of the prepared cavity. Generally, a gap of 10–15 µm exists around the restoration. However, the adaptation to the cavity wall improves with time; referred to as '*self-sealing restoration*'. This is by virtue of accumulation of corrosion products in the interface with the passage of time. The corrosion products are the oxides and chlorides of tin, mostly occurring in the low copper alloys. However, high copper alloys confer a greater resistance to corrosion. Because of this increased resistance and a slower rate of forming corrosion products, microleakage in the high copper restorations continues for a longer period of time.

Microleakage, as a result of dimensional changes in amalgam, is minimal. During setting of the conventional mercury/alloy mix, a small contraction is seen initially, when the mercury is consumed. This is followed by a small expansion as the crystalline matrix is formed. Modern alloys including high copper alloys undergo very little net dimensional change (±0.2% by volume) (*ADA specification No.1 allows dimensional change of 20 µm/cm for a set amalgam*).

Measures to Reduce Microleakage

i. *Different types of silver alloys* exhibit different levels of microleakage. Spherical alloys display more microleakage as compared to lathecut/admixed alloys (Spherical alloys do not adapt properly and show shrinkage during setting).
ii. *Condensation of amalgam* should be carried out immediately after trituration. A longer time lapse results in the loss of plasticity and increases the chances of internal voids. Insertion of amalgam should be carried out in increments. Adequate condensation removes voids and

promotes adaptation of the material to the cavity walls. Condensation should first be initiated in the center, then moving towards the periphery, using the stepping process.
Mechanical (pneumatic) means of condensation is considered superior to hand condensation in controlling marginal leakage.

iii. *Burnishing of amalgam* reduces microleakage by aiding adaptation of the material and enhancing homogeneity. The effect of burnishing varies with the particle shape of amalgam. With spherical alloys, no reduction of leakage results from the burnishing procedure as compared to lathe cut and admixed alloys (The spherical shape of the alloy allows the particles to be 'pushed aside' during the process).

iv. *Alloys with lesser creep values* show a smaller degree of marginal leakage. A creep rate of less than 3.0% is considered acceptable (ADA specification No. 1). Low copper alloys show a creep value ranging from 0.8–8.0% as compared to high copper alloys, which have much lower values of 0.1–1%.

v. *Sealing the cavity walls with varnish* under an amalgam restoration prevents microleakage at least until the corrosion products of amalgam begin to fill the microspace. Since varnish is soluble in the oral environment, its effectiveness is limited. The time lapse, however, is sufficient for the corrosion products to replace the varnish and permanently seal the interface. Application of a varnish layer under high copper amalgams did not effectively reduce the microleakage.

vi. *Applying coating* of unfilled resin over the restoration margins and the adjacent enamel effectively reduce microleakage. Although the resin may finally wear away, it delays microleakage until the corrosion products are accumulated in the interface.

vii. *Coefficient of thermal expansion* of amalgam (25×10^{-6}/°C) is very much different from that of the tooth structure (11.4×10^{-6}/°C). Hence, the change in dimensions of the gap because of thermal changes is of concern for amalgam (corrosion products do help closing the gap).

viii. *Bonding of amalgam to enamel and dentin* have proved effective in reducing the microleakage.

ix. *Gallium alloys* [alloys of gallium-tin-indium (liquid) and a silver based alloy] having high wetting ability produce restoration that is highly resistant to microleakage. They have such a high wetting characteristic that when the gallium-rich mixture is inserted into the cavity, some of it tends to move up along the sides of the instrument.

b. Microleakage around Glass-Ionomer Restorations

Glass-ionomer cement adheres to the tooth structure by forming chemical bond between carboxyl groups of the cement and calcium ions of the tooth. Since glass-ionomer is hydrophilic in nature, it can bond even in the presence of a slightly wet surface. Other properties, which contribute to minimize leakage, are: (i) A coefficient of thermal expansion, closely matches to tooth structure, (ii) fluoride releases into the surrounding environment, and (iii) the ability to renew broken ionic bonds.

Glass-ionomer cement is highly technique sensitive; isolation from moisture, at least 30 minutes after placement is important. On exposure to water, the matrix forming ions are easily leached out during the initial set, which could interfere at tooth–restoration interface. Solubility in oral fluids continues to be a problem for 24 hours following insertion of the material. Excessive dehydration/desiccation can result in a crazed/cracked surface, which if extends to the margins, could result in considerable marginal leakage. Coating the restoration with surface protective agents, like cavity varnishes or unfilled resins, is recommended for protecting against dehydration. Once the cement has attained maturity, it becomes resistant to solubility and disintegration in the oral fluids. Another factor that may harm the marginal integrity of the restoration is the use of sharp hand instruments for finishing, before the material has completely set.

Measures to Reduce Microleakage

i. Proper manipulation and placement of glass-ionomer cement on clean dentinal surfaces (Lower powder: liquid ratio may increase the solubility of the cement).

ii. Rotary instruments are preferred for finishing restoration margins. Hand instruments may tear away the material at the margins resulting in marginal ditching.

iii. During finishing, restoration surfaces should be coated with vaseline or petroleum jelly to protect from moisture contamination and desiccation.

iv. Two coats of varnish or a layer of unfilled resin should be applied over the restoration. Varnish is semipermeable in nature as compared to unfilled resin, which is more resistant to water contamination. Despite of this fact, a varnish application is preferred as it closely adheres to the surrounding surface of enamel, preventing any moisture from reaching the glass–ionomer–tooth interface.

v. Prior conditioning of the tooth surface before insertion of the cement increases the bonding and reduces the microleakage. Various conditioning

agents advocated for the purpose are, tannic acid, polyacrylic acid, citric acid, etc.

Resin-modified glass-ionomer cements, because of rapid initial setting, reduce the effect of moisture contamination. Unfortunately, resin-modified glass-ionomers have occasionally shown increased levels of microleakage as compared to the chemically cured ones. This is probably because the resin components in the former lead to shrinkage during polymerization. In contrast to the slow setting chemically cure glass-ionomer cements, which permit stress relief, the resin-modified systems exhibit more rapid setting contraction through light polymerization. Moreover, less water and less carboxylic acid content also decrease the wetting ability of light cure glass-ionomer cements to the tooth substance promoting marginal leakage. To a certain extent, expansion due to water uptake, which is a function of the resin component of cement, compensates for the initial resin shrinkage and reduces microleakage.

c. Microleakage around Composite Restorations

Composite resins do not possess the quality of bonding to either enamel or dentin. The inability to bond, coupled with marked polymerization shrinkage and thermal expansion/contraction predispose to marginal leakage. Prior treatment of the tooth surface, like acid etching/conditioning and the use of bonding agents, facilitates margin adaptability for composite restorations. In enamel, such treatment leads to an unparalleled success in preventing microleakage. Conversely, the bond to dentin, though substantially reduces microleakage, but does not protect the dentin–restoration interface completely. The adequate adhesion is limited by a number of factors, like vitality of the dentin, difference in physical and chemical composition of dentin as compared to enamel, presence of dentinal fluid, smear layer, etc. Development of internal stresses from polymerization shrinkage and thermal effects also have a detrimental effect on the bond.

The most notable factors that contribute to marginal leakage of composites are:

1. Composite restorations are technique sensitive. Any inadvertent step can lead to failure, including an increase in marginal leakage.
2. Marginal gaps at the tooth–restoration interface are primarily due to polymerization shrinkage of the setting resin. After the setting, the gaps are further affected by masticatory forces, thermal changes and water sorption.

   The features affecting gap formation are:

   i. The volumetric polymerization shrinkage in composite restorations is in the range of 1.67–5.68% (the lesser being for the light-activated ones).

   In case of using bonding agents, shrinkage results in the development of tensile/shear stresses at the tooth–restoration interface. Within certain limits, the adhesive bond is able to withstand these stresses. Once the stresses exceed the bond strength and the plastic or elastic deformation of the combined system, a separation at the tooth–restoration interface may occur leading to microleakage. It is usually the bond with dentin which is compromised during shrinkage; whereas, the bond with enamel is sufficiently strong to withstand the same amount of forces.

   ii. Functional occlusal stresses may enhance because of repeated plastic or elastic deformation of the restoration.

   iii. The marked difference in the coefficient of thermal expansion of restorative resins (22–25 $\times 10^{-6}$/°C) and tooth structure (11.4 $\times 10^{-6}$/°C) has a detrimental effect on adhesion.

   iv. Composite resin restorations have a tendency to absorb water from the environment, causing the restoration to expand (water sorption counteract polymerization shrinkage to a little extent). In contrast to the polymerization contraction stresses, which are generated at a rapid rate, relief of stresses by hygroscopic expansion proceeds slowly. The resins with the largest quantity of filler have the least water sorption (Water uptake is the property of the resin component of the matrix). Though water sorption may improve marginal adaptation of composite resin, it impairs its mechanical properties.

   v. Marginal gaps are higher on the cervical margins of a Class II restoration; reasons being:
   - Placing the restorative material at these areas is difficult.
   - Entrapment of air during placement.
   - Difficulty during condensation because of the pull back of sticky material.
   - Inadequate bonding to the gingival wall due to polymerization shrinkage.

Measures to Reduce Marginal Leakage

A number of measures have been advocated to enhance marginal adaptation and reduce microleakage of composite restorations.

The advocated measures are as follows:

i. ***Choice of composite material:*** Microfilled composite resins provide a better marginal adaptation over the macrofilled composite resins, because of:

- Greater flexibility of the microfills during polymerization shrinkage decreases the contraction forces that tend to weaken the dentinal bond.
- Larger absorption of water resulting in an expansion, which counteracts polymerization shrinkage to some extent.

The amount of polymerization shrinkage can be markedly controlled, if the resin to filler ratio is decreased. This phenomenon led to the introduction of beta quartz glass inserts, used as '*mega fillers*' in composite restorations. Inserts are made of lithium aluminosilicate glass, and are available in various sizes and shapes. Inserts tend to increase the filler content relative to the amount of resin, decreasing the contraction. They also have a low coefficient of thermal expansion ($4\times 10^{-6}/°C$); exhibit no polymerization shrinkage and water sorption.

Prepolymerized composite balls, substituted for the glass inserts, however, showed significantly greater microleakage.

ii. ***Cavity design:*** Cavity design for composites should be as conservative as possible. Being adhesive to the underlying tooth structure, retentive and extensive preparations are not required. Placement of bevels, reduced depths and rounded internal angles are effective in providing good marginal adaptation and reducing leakage (Fig. 22.6).

Not only the size, but the shape of the cavity is also an important factor in determining the width of the marginal gap. This is illustrated in Fig. 22.7, which shows three dentin cavities of identical diameter but of different shapes. The cavity to the right is V shaped and the cavity to the left is box shaped. When the cavities are filled with composite resin without any bonding agent, the marginal gap is reduced from 13 m to 4.0 m from left to right; whereas, when dentin bonding system is applied, the gap width is in the order of 3.0 m



Fig. 22.6: Class I cavity preparation for composite in mandibular molar

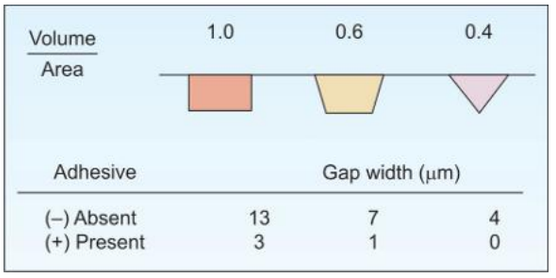


Fig. 22.7: Width of the marginal gap in relation to the shape of the cavity for composite

to almost nil. This can be explained on the decreasing ratio of volume/bonded area from 1.0 for the box-shaped cavity to 0.4 for the V-shaped cavity.

The role of bevels on cavosurface margins in reducing leakage remained controversial. The principle favoring bevels is to increase the surface area for enamel bonding and reduces microleakage. Occlusal beveling is usually not required because of the enamel rod direction and the possibility of extending the material to load-bearing areas. Beveling of cervical margins in extensive Class II preparations may result in removing the remaining enamel. Placement of bevels is recommended mainly on the accessible facial and lingual margins of the proximal box.

iii. ***Acid etching and bonding:*** Acid etching removes surface contaminants, raises the surface energy and reactivity of enamel and increases the surface area for bonding. The bonding agent is drawn by capillary attraction into the microporosities created by acid etching. These polymer tags provide micromechanical interlocking with enamel, thereby reducing microleakage to almost nil.

A shortcoming with this technique is its dependence on the presence of thick enamel at the cavity walls. The effectiveness of marginal seal is compromised by the position and surface structure of enamel as it anatomically varies in different areas of the same tooth. The results are obvious when the restoration margins are located in the cervical enamel. Cervical enamel is thin, irregular in prism structure and devoid of characteristic prism markings. Therefore, the bond to cervical enamel is not very intimate as compared to the occlusal enamel.

Acid etching on dentin was traditionally discouraged because etchants were thought to open and widen the dentinal tubules resulting in increased permeability and bacterial ingress to the underlying pulp. However, with the introduction of the newer hydrophilic resins, the concept has

changed. Many authors propose deliberate etching of dentin to create open tubules and a porous intertubular layer. It is believed that a close adhesion can be attained between resin adhesive and the treated dentin.

New bonding agents based on glass-ionomer technology have been developed (e.g. Scotchbond Multipurpose). These have carboxylic acid groups, which facilitate attachment to dentin. They are also useful in attaching composite resins to glass-ionomer cement surfaces. These agents can be separately grouped as '*glass-ionomer bonding agents*'.

All newer bonding agents (composition may vary) facilitate close adaptation of restorative material to the tooth structure.

iv. ***Restorative procedure:*** Placing composite resin in small increments is recommended to control polymerization shrinkage and the contraction gap at the margins (Bulk placement shows poor adaptation). Thickness of one increment should preferably be limited to 1.0–1.5 mm. Different incremental placement designs have been described in Chapter 19.

The concept of *soft start polymerization* has been advocated to reduce marginal gap and improve marginal integrity. The procedure involves a short prepolymerization at low intensity followed by the final cure at high intensity. This allows composite resin to flow during the initial setting period, thereby minimizing the stresses induced at the interface. Prolonging the curing time of a composite resin improves upon the microleakage by allowing a greater inversion rate of the monomer component.

v. ***Direction of the light source:*** It is hypothesized that the polymerization shrinkage of setting composite resin is directed towards the light source. Because of the inadequate access of the light source in proximal areas and also since the curing tip is usually directed from the occlusal aspect, the gingival increment shrinks occlusally. Use of curing aids like light directing wedges, flexible light guides and focusing tips can facilitate better curing in areas of poor access. A three-sited light curing technique for better adaptation of conventional Class II composite restorations has been advocated (Fig. 22.8).

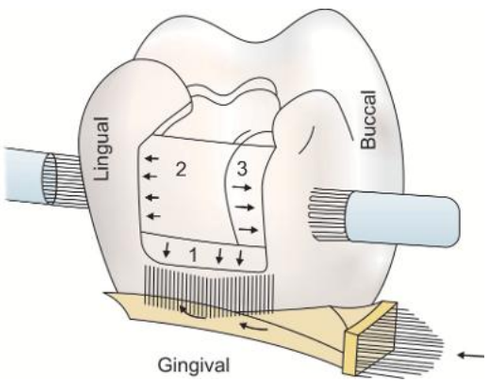


Fig. 22.8: Three-sited light curing technique

vi. ***Sealing the marginal gaps:*** After finishing and polishing of composite restoration, an unfilled low viscosity resin is applied to the margins by slowly brushing the resin laterally from one side to another. Materials used for the purpose of sealing gaps are the commercially available pit and fissure sealants. Sealing marginal gaps has shown promising results in reducing marginal leakage.

vii. ***Finishing process:*** Finishing techniques and their timings affect the ability of restorative materials to resist leakage. Most of the authors advocate finishing of composites to be delayed for 24 hours, until the polymerization is complete. If done before the initial setting, it may break the bond between resin and the tooth. Finishing after one day was considered acceptable as it resulted in almost no microleakage. However, the delayed finishing after one week can be detrimental to the already formed marginal seal of the restorations. Stresses, which disrupt the adhesive bonds, are always generated during the finishing process.

Variations in the finishing techniques does affect leakage. Dry finishing techniques lead to increased leakage. It is, therefore, recommended that rotary instruments should be used at slow speeds using light intermittent strokes and with generous air coolants. Use of Soflex discs for contouring and finishing has shown the best marginal quality for composite restorations.

viii. ***Use of sandwich technique:*** Glass-ionomer cements under composite resins have been successful in reducing microleakage. Bond of glass-ionomer cement to enamel is weaker as compared to composites; however, the bond to dentin is stronger. The bond between etched glass-ionomer and composite is also stronger. It shows a chemical union with dentin even in the presence of dentinal fluid because of its hydrophilic nature, possesses a coefficient of thermal expansion close to that of tooth structure, acts as a fluoride reservoir and is kind to the pulp. All these properties make glass-ionomer cements an excellent lining material under the composite restoration. This is the basis for bilayered restorations, previously known as 'sandwich restorations.' However, it is believed that the polymerization shrinkage of composite resin breaks the seal of glass-ionomer cement to

dentin. The contraction tends to pull the base away from the cavity floor, resulting in a gap not less than 5.0 mm at the cavity–restoration interface. The created gap has a potential for leakage and fluid percolation. Light cure glass-ionomer, when used as base, shows better performance as compared to chemically cure counterparts. This might be because of their ability to adhere immediately to dentin and achieve high early tensile and compressive strengths. The hardened cement resists stresses that tend to disrupt the dentinal seal in a better way.

ix. ***Use of composite inlay restorations:*** Composite inlay restorations bond to the tooth structure using composite as a cementing medium. Light cure composite luting agents are not advocated, as it is difficult to cure them through a pre-processed resin restoration. Composite inlay seems to be an efficient way of managing polymerization shrinkage since the bulk of the contraction occurs prior to cementation. Less leakage with inlay systems as compared to direct resin restorations has been established.

Luting a composite inlay is always challenging. Since the luting cements are less filled than the resin restorative materials, they tend to wear off easily. The hybrid luting resins are more susceptible to wear than microfilled ones. Hence, microfilled resins should be preferred for the cementation purposes.

Another drawback is that the cement often fails to bond chemically with the inlay. This is attributed to the curing procedure, which induces a high conversion rate of the inlay and greatly reduces the availability of remaining unconverted monomers for copolymerization with the resin lute. A passage for microleakage is created, resulting in marginal staining. For improving the bond between composite inlay and the resin cements, different methods have been advocated, like (i) use of a solvent, such as ethyl acetate to soften the cavity side of the inlay prior to cementation, (ii) sand blasting the cavity surface of the inlay with aluminum oxide abrasive particles, so as to increase the surface area for bonding, and (iii) etching the cavity side of the composite inlay with 10% hydrofluoric acid.

x. ***Expanding matrix resins for dental composites:*** Composite resins that expand during polymerization can be helpful in reducing the leakage. These would facilitate bulk placement of the material, reduce postoperative sensitivity, secondary caries and microleakage.

d. Microleakage around Direct Gold Restorations

Direct gold restorative materials efficiently adapt to the cavity walls. The good adaptability of gold to the cavity walls is because of inherent qualities as:

- High malleability and ductility of gold facilitates burnishing (riding the gold over the marginal voids); minimizing leakage.
- A short bevel on the cavosurface margins facilitates the burnishing, contouring and polishing thereby promoting close adaptation at the cavity–restoration interface.
- Insolubility in the oral fluids.
- Condensation of gold induces elastic compression of the underlying and surrounding dentin adapting the material adequately to the cavity walls.

Measures to Reduce Leakage

i. In general, mat and powdered golds are porous and less effective than cohesive gold foils in obtaining a good cavity seal (Former should be used for forming the internal bulk of the restoration and the latter be used as a veneer on the surface to prevent leakage).
ii. Uniform stepping of the condenser point is indicated to drive away all air spaces and closely adapt the material to the underlying surface (Stepping should proceed from the center to the periphery).
iii. Lines of force should be directed at 90° to the pulpal floor in the center of the cavity preparation and then changed to 45° to the cavity walls at the periphery.
iv. Average force of 10 pounds applied with a 1.0 mm condenser point is considered optimal for proper adaptation of the material to the cavity walls and cohesion of two gold surfaces.
v. Surface procedures, like burnishing, finishing and polishing bring the metal closer to the tooth structure improving the marginal seal.

e. Microleakage around Cast Restorations

The cast restorations do not offer close adaptation to the cavity walls. A gap ranging from 10 to 160 μm has been reported in various cast restorations. An intermediate layer of luting cement is necessary to seal the interface. The low viscosity luting cements are able to penetrate the irregularities on both the tooth and the restoration surface creating micromechanical retention. With the advent of adhesive cements, chemical retention is also possible.

The intermediate cement layer, used for sealing intervening gaps, may promote leakage. Though the recent cements are less soluble in oral fluids (Zinc phosphate cement earlier used with cast gold

restorations was highly soluble), yet none of the materials totally resists the solubility in oral fluids. Type II and type III gold alloys allow a percentage elongation of 20–35%. For the highly soluble cements, like zinc phosphate, burnishing should be delayed for 24 hours. This gap of time period allows for the superficial a few microns of the cement to dissolve leaving behind metal margins that can be burnished. For the non-soluble resin cements, like resin lutes, burnishing can be performed immediately. Marginal gaps, which exceed 0.1 mm, should not be burnished; rather the castings remade.

Leakage continues to be a problem in areas where the cement line is inadvertently exposed to the oral environment. This condition exists when the margins are not adequately beveled and/or burnished, e.g. in regions difficult to be reached by the burnisher, like the gingival margins of a Class II inlay, or when harder gold used for crowns is not easily burnished.

Measures to Reduce Microleakage

i. Cavosurface margins should be beveled in cast gold restorations
ii. Resin-based luting cement should be preferred, being less soluble in oral environment.
iii. Lining of glass-cermat can be placed at the gingival wall of Class II preparation (area more prone to leakage).

f. Microleakage around Porcelain Restorations

Dental porcelain is a brittle material with low tensile strength; requires adequate bonding to the underlying tooth surface for gaining structural and functional integrity. Porcelain restorations are luted with adhesive resin cements. The resin cements should preferably be dual cure to allow polymerization to be completed in areas inacessible to light.

The surface of the inlay is treated chemically to achieve an adequate bond of the ceramic inlay and the resin luting agent. Acid etching of the inlay surface is carried out using hydrofluoric acid (fired porcelain) and ammonium bifluoride (milled and cast ceramics). The etched surface is then silanated to promote wetting with the resin, improving chemical retention. However, the chemical bond between silane and luting cement improves only the initial bond strength. Later on, the bond gets weakened, probably by hydrolysis, which leads to marked decrease in bond strength after one year. Resin luting cements should not be applied without prior tooth bonding procedures because significant microleakage could occur due to inherent polymerization shrinkage.

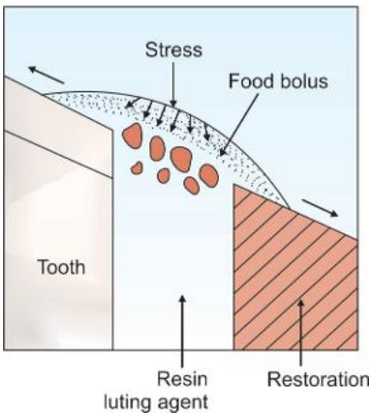


Fig. 22.9: Diagram showing wear related to resin lute

The most vulnerable site of leakage continues to be the wear of cement lute and the interface within inlay and the tooth (Fig. 22.9).

Measures to Reduce Microleakage

i. The inner surface of the preparation is etched for better adhesion.
ii. Appropriate bonding agents and resin-based luting agents should be used.

METHODS TO DETECT MICROLEAKAGE

Various methods demonstrating microleakage have been applied to restorative materials. Most of these methods are employed *in vitro*. Though *in vitro* tests try to simulate oral conditions by thermocycling, etc., yet the dynamic nature of pulpo-dentinal complex and its defense mechanisms cannot be easily simulated *in vitro*. More so, the accumulation of plaque and other agents might vary the microleakage results *in vivo*.

The methods employed to detect microleakage are:

1. Penetration methods
   a. Dye penetration
   b. Chemical tracers
   c. Radioactive isotopes
   d. Neutron activation analysis
   e. Bacterial penetration
2. Air pressure technique
3. Fluid filtration method
4. Electrochemical method
5. Microscopic examination
   a. Scanning electron microscopy
   b. Fluorescent microscopy
   c. Optical coherence tomography
6. Miscellaneous
   a. Artificial caries
   b. Pain perception
   c. Reverse diffusion method
   d. Three-dimensional computer analysis



Fig. 22.10A: Detection of marginal leakage (cervical wall)

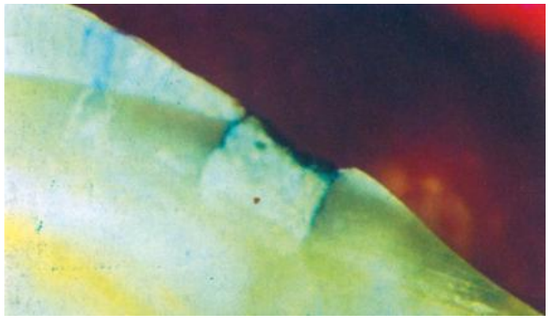


Fig. 22.10B: Detection of marginal leakage (occlusal and cervical walls)

e. Laser microspectral analysis
f. Resin infiltration method

1. Penetration Methods

a. *Dye Penetration*

The use of organic dyes is one of the oldest and most commonly used methods for detection of microleakage. The dye has a contrasting color vis-à-vis the tooth and the restoration (Fig. 22.10A, B). Agents most frequently used for this purpose are: Methylene blue, India ink, crystal violet, fluorescein, rhodamine B, eosin, basic fuchsin, erythrosin, etc. These dyes are available either as solutions or particle suspensions of different particle sizes. They should not bond either with the tooth substance or the restoration and should be color stable under all conditions of investigation.

The technique involves immersion of extracted and restored tooth in a dye solution for a predetermined period. After the stipulated time, the tooth is removed, washed, sectioned and examined under a microscope to establish the extent of penetration of dye. The results are quantified by assigning numbers to the defined depths of penetration.

The validity of using dyes is, however, questioned because of the reasons, such as:

- With a variety of dyes having different particle sizes, it cannot be expected that consistent results will be obtained even if standardized techniques are used. It has been shown that different concentrations of two dyes can vary in penetration times from 5 minutes to 1 hour.
- A potential source of error exists when the dye binds to the tooth substance or restorative material, e.g. basic fuchsin dyes have shown to preferentially bind with the carious dentin. Therefore, when attempting to demonstrate a space between the tooth and restoration, an area of stained carious dentin can potentially be mistaken for a larger gap than actually exists.
- It is recommended that dye penetration studies should be carried out in vacuum; however, it is difficult to eliminate trapped air in small spaces.
- Dyes that are not color stable may lead to misinterpretation of results. For example, aniline blue becomes colorless in alkaline conditions, such as in presence of calcium hydroxide.

The assessment of results is only subjective, which is a major disadvantage with this method.

b. *Chemical Tracers*

Chemical tracers rely upon the reaction between one or more chemicals used. The technique involves reaction of two colorless compounds to produce an opaque precipitate, usually a *silver salt*. It is essential that both the chemicals penetrate the margin, (Precipitation does not occur, if only one of the two molecules penetrates.)

The procedure followed is the same, i.e. extracted tooth with restoration is immersed in a 50% silver nitrate solution, which is subsequently reacted with photographic developer (benzene hydroquinone). 1.0% silver chloride has also been used. The limitations of chemical tracers are similar to dye penetration studies.

c. *Radioactive Isotopes*

Radioactive isotopes, such as ^{45}Ca, ^{131}I, ^{32}P, ^{14}C, ^{35}S, ^{86}Rb, etc. are used in a manner similar to that of dyes for assessing microleakage. The specimens are autoradiographed to detect the tracer. The isotope can detect even minute amount of marginal leakage, since the size of the isotope molecules is only 40 nm compared to the smallest size of the dye particles (120 nm).

Limitations of this technique are:

- Subjective assessment of results; however, with the use of stereomicroscope, the subjectivity can be minimized.
- A high energy isotope produces scattering on the film leading to an increase in apparent leakage. Hence, isotopes with low energies should be preferred for resolution.

- Isotopes, such as ^{45}Ca, which have an affinity for the tooth or restorative material, may lead to misleading results.
- The isotopes are comparatively expensive and technique sensitive. All precautionary measures are to be taken to satisfy safety requirements.

d. Neutron Activation Analysis

The restored tooth is soaked in an aqueous solution of a non-radioactive manganese salt. The tooth is then placed in the core of a nuclear reactor, where bombardment with neutrons activates the ^{55}Mn to ^{56}Mn. The radiation subsequently emitted by the tooth is measured to quantify the volume of the tracer present.

Limitations of this technique are:

- Inability to identify the points where the restoration has leaked.
- A combined effort is mandatory from nuclear engineers and the operator.
- Manganese may be absorbed by the tooth or restorative material.
- Heavy costs.

e. Bacterial Penetration

The possibility of bacteria penetrating through or around the filling material is the base of this method. The restored teeth are immersed in cultured broths; the filling removed and the dentin shavings from the base of the cavity cultured. The disadvantage with this method is that the results are qualitative and not quantitative. Moreover, marginal gaps that allow passage of bacteria are usually 0.5–1.0 µm. These techniques are not valid for the gaps smaller than this size, which could allow penetration of harmful toxins.

2. Air Pressure Technique

The air pressure technique (device used to test marginal leakage) is shown diagrammatically in Fig. 22.11.

Compressed air is introduced towards the tooth-restoration interface. The release of air bubbles at the margins provides a subjective view of marginal integrity.

Advantages

- Tooth need not be destroyed.
- Result can be quantified.

Disadvantages

- Inability to use *in-vivo*
- Does not represent the clinical situation and also does not take into account the drying effect of compressed air.
- It is possible that some air may leak before it enters the tooth.

3. Fluid Filtration Method

The pressurized-liquid is applied onto the restored tooth with a constant pressure generated by a gas system. The teeth are sectioned and connected to a plastic tube, filled with de-ionized water. A standard glass capillary tube is connected to the plastic tube at the outlet of the specimen. The water is sucked back from the capillary tube creating air bubbles. The whole set-up is placed in water bath at a constant temperature. An adjusted pressure is applied from the inlet side to force the water through the void along the filling, thereby displacing the air bubble in the capillary tube. The volume of fluid transported is measured by observing the movement of the air bubble (Movement of air bubbles are best measured by computer-controlled diode laser beam) (Fig. 22.12).

Advantages

- Samples are not destroyed.
- Results are automatically recorded; no bias.
- Accurate results, because very small volume can be recorded.
- No need of any tracer; overcomes the problem of molecular size.

Disadvantage

- Standardization of pressure, vis-a-vis, movement is difficult.

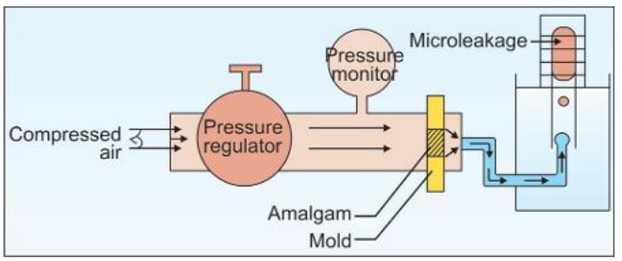


Fig. 22.11: Air pressure technique: Device used to measure microleakage at the amalgam–mold (tooth–restoration) interface

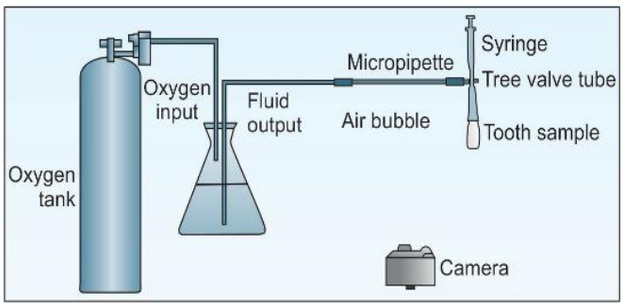


Fig. 22.12: Fluid filtration method

4. Electrochemical Method

The method involves the insertion of an electrode into the extracted tooth in such a way that it contacts the base of the restoration. The tooth is suitably sealed to prevent any electrical leakage through natural tooth structure and then immersed in an electrolytic bath. A potential is applied between the tooth and the bath, and the leakage is assessed by measuring the current flowing across the interface.

Drawbacks

- Not suitable for metallic restorations
- Inability to use *in vivo*.

5. Microscopic Examination

Sequential microscopic sections are required to study dental tissues in three dimensions. Routine sectioning is time consuming and may lead to wastage. In order to overcome these limitations, *acetate peel technique* is used, wherein tooth tissue is ground with carborundum powder on a glass plate. A thin layer so produced is dried and mounted for examination under microscope.

a. *Scanning Electron Microscopy*

The scanning electron microscope (SEM) provides direct visual observation of the tooth–restoration interface because of its high magnification and depth. The low vacuum SEM evaluates the rubber base impression directly, decreasing the number of steps and the possible chances of inaccuracy.

Limitation

- Potential to induce artefacts during specimen preparations.

b. *Fluorescent Microscopy*

Fluorescent microscopy uses UV radiation on selected dye, which is capable of absorbing these radiations at one wavelength and emitting at different wavelength.

Fluorescent dyes are detectable in dilute concentrations sensitive to UV light, easy to photograph, permitted more reproducible results and not expensive. The contrast of the natural fluorescence of the tooth against that of dye provided a contrast that made it easy to detect the path of dye penetration under UV light.

c. *Optical Coherence Tomography (OCT)*

Optical coherence tomography (OCT) has successfully evaluated restoration margins and cracks at the margins (width and depth of the gaps). The amount of reflection of infrared light from the margin gaps can be quantified.

6. Miscellaneous

a. *Artificial Caries*

Artificial caries like lesion is produced *in vitro* using bacterial cultures. The lesions produced along the cavity walls and under the cavity walls are assessed. The histological appearance of such lesions under polarized light shows two parts: An 'outer lesion' and a 'cavity wall' lesion. The outer lesion exhibits the features of primary attack, while the cavity wall lesion was formed by microleakage of ions from the caries-inducing medium into the tooth–restoration interface.

Advantages

- Microleakage can be directly related to the development of artificial lesion.
- Quantification of results is possible using depth of the lesion as the measurable parameter (If degree of demineralization is used as a measure, it provides only subjective assessment).

b. *Pain Perception*

The suspected margins of the restorations are painted with calcium chloride and kept there for a few minutes. If there is an open communication to exposed dentin from tooth–restoration interface, the patient will complain of pain within 30–60 seconds. The response may be delayed because it takes time for the fluid to move osmotically through that space.

Disadvantage

- Subjectivity regarding the pain perception of the patient.

c. *Reverse Diffusion Method*

The reverse diffusion method involves placing a certain amount of tracers on the cavity floor prior to insertion of the restorative material. After a stipulated period of time, the tooth is immersed in a definite volume of the medium. The amount of the tracers which leaks into the medium is measured (Fig. 22.13). This method is also modified by using calcium hydroxide in place of tracers at the base of the cavity. The pH of calcium hydroxide is evaluated on litmus

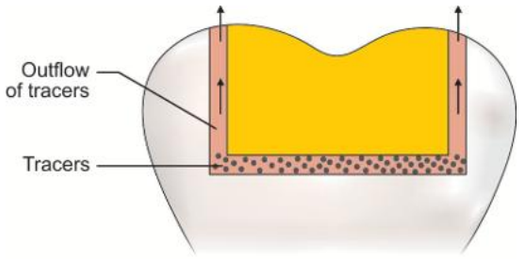


Fig. 22.13: Reverse diffusion method

papers, placed at the tooth–restoration interface. Reverse diffusion method is considered a better method as it reveals time-dependent leakage at any given time.

It is suggested that the 'microleakage coefficient' should be considered, which take into account all factors including marginal gaps during diffusion process.

Limitations

- Minimum amount of tracer is necessary at a given time.
- How to quantify marginal leakage with the numbers of tracers?

d. *Three-dimensional Computer Analysis*

The three-dimensional analysis utilizes serial sections (approximately 200 µm thick and separated by 280 µm). Three-dimensional models are created and reconstructed by computer-aided tools. Computer image analyzer is applied to count the surface areas of dye leakage; however, the volume of leakage is calculated manually. It has been observed that microleakage in the three-dimensional analysis was significantly better analysed than the conventional method.

Currently, CBCT is employed for three-dimensional analysis of microleakage.

e. *Laser Microspectral Analysis*

Laser microspectral device consists of an infrared pulsed laser, usually with ruby or neodymium doped glass as active medium. The energy of the laser used is around 0.5–2 J/pulse, which is sufficient to produce a crater of 20–300 µm diameter on the surface of the probe. Vapors ejected from the crater having high temperature (20000°K), initiate an electric discharge between the electrodes. The radiation produced by the discharge is focused to the spectrophotometer, which provides emission spectrum of the atoms; subsequently, measuring the infiltration of dye in the tooth–restoration interface.

f. *Resin Infiltration Method*

The specimens are coated with nail polish within 2.0 mm of the end of apex. The resin is prepared by dissolving 1.4 gm of resorcinol in 2.0 mL of 40% formaldehyde solution and pH is adjusted to 8.2 with aqueous potassium hydroxide immediately before use. The teeth are immersed in resorcinol formaldehyde resin of 5 days at 4.0°C and resin is allowed to polymerize for 4 days. They are then embedded in epoxy resin and sectioned horizontally. The cross-sections are transilluminated and viewed at magnification of 25× under stereomicroscope. The photographs are evaluated for image analysis.

NANOLEAKAGE

Nanoleakage implies leakage around the collagen fibrils within the hybrid layer that have not been completely infiltrated by resin. Leakage may be seen at the bottom of the hybrid layer and/or scattered along its entire width, depending upon the bonding system employed. It appears that bonding systems that etch deeper into dentin are likely to show higher degrees of nanoleakage. This could be the possible reason for reduced degree of leakage observed with self-etching/self-priming system than with systems that use an acidic conditioner as a separate step. The clinical significance of nanoleakage indicates the inability of the adhesive resin to completely infiltrate the demineralized dentin, thereby leaving behind pores and spaces. Penetration pathways in nanoleakage are porosities in a range less than 50 µm. Poorly infiltrated hybrid layer contains voids that predispose to accumulation of water and oral fluids, which may accelerate the degradation of the bond. When demineralized dentin is infiltrated with resin, its modulus of elasticity is much higher than the demineralized dentin. If the hybrid layer fails to be penetrated completely, its modulus of elasticity might fall in between that of adhesive resin and the un-infiltrated dentin collagen. Contact with water and oral fluids would provide a gradual decrease in the modulus of elasticity of the bonded interface reducing the stress concentration in this area. Due to slow hydrolysis, the integrity is not preserved for a longer periods. Therefore, dentin bonding agents, may reduce microleakage initially, their long-term performance under stress and continued exposure to oral fluids is questionable.

The techniques used to determine nanoleakage are as follows.

1. Tracer (Dyes)

Several silver dyes have been used to test the sealing ability of dentin adhesives; such as, silver nitrate, silver methenamine and ammonical silver nitrate. It is shown that the hybrid layer is somewhat porous and is accessible to a silver dye even in the absence of marginal gap. 50% aqueous solution of silver nitrate is commonly used. Nanoleakage is a significant problem even with the newer adhesives. A few authors have confirmed that the penetration of tracer is because of incomplete resin penetration within the hybrid layers; leakage may be because of increased porosities and high hydrophilicity of the adhesives. The tracer dyes are analyzed within the hybrid layer using the microscopes.

2. Scanning Electron Microscopy (SEM)

The scanning electron microsocpy (SEM) provides a mean of direct visual observation of the hybrid layer because of its high magnification and depth of focus. Even the submicron defects can be observed at required magnification, such as 200× or 1000×.

Limitations

- Drying of specimens, dehydration and embedding, which is needed for SEM analysis may result in artefacts.
- Cutting/breaking the sample may influence the surface morphology.

3. Transmission Electron Microscopy (TEM)

Transmission electron microscopy (TEM) is an imaging technique, whereby a beam of electrons is transmitted through a specimen; the image so formed is magnified and directed to appear either on a fluorescent screen or layer of photographic film, or a CCD sensor. TEM can resolve object details smaller than 1.0 nm, effectively analyse nanoleakage.

4. Field Emission Scanning Electron Microscopy (FESEM)

A field-emission scanning electron microscope provides narrower probing beams, resulting in improved spatial resolution and minimal sample damage. FESEM produces clearer, less electrostatically distorted images with spatial resolution down to 1.5 nm, i.e. 3 to 6 times better than conventional SEM.

5. Confocal Laser Scanning Electron Microscopy (CLSEM)

Confocal laser scanning electron microscopy (CLSEM) merges the capabilities of the scanning, transmission and X-ray microscopes, and achieves unprecedented resolutions in optically dense materials, by implementing the technology of confocal imaging into an electron microscope.

CLSEM has also been used in combination with fluorescent dye (rhodamine B) to determine nanoleakage. The advantages of CLSEM are that, it is a non-destructive examination and drying of the samples is not necessary.

6. Coherent Diffraction Imaging (CDI)

Coherent diffraction imaging (CDI) utilizes a highly coherent beam of electrons incident on a sample to generate a diffraction pattern. This pattern is recorded electronically and then used to reconstruct an image. The resolution is limited to about 10 nm for frozen-hydrated samples, but resolutions of as high as 1.0 to 2.0 nm are possible for inorganic materials less sensitive to damage. It has been proposed that radiation damage may be avoided by using ultrashort pulses of X-rays, where the time scale of the destruction mechanism is longer than the pulse duration. The advantage of CDI is that the final image is aberration free.

BIBLIOGRAPHY

1. Alani AH, Toh CG. Detection of microleakage around dental restorations: A review Oper Dent 1997;22:173–78.
2. AlHabdan AA. Review of microleakage evaluation tools. J Int Oral Health 2017;9:141–45.
3. Alvarenga FA, Andrade MF, Pinelli C, Rastelli AN, Victorino KR, Loffredo LD. Accuracy of digital images in the detection of marginal microleakage: An in vitro study. J Adhes Dent 2012;14:335–8.
4. Amarante DCDA, Sinhoretti MA, Correr SL, Consani S. Influence of methodology and evaluation criteria on determining microleakage in dentin-restorative interfaces. Clin Oral Invest 2006; pp 1–6.
5. Bakhsh TA, Al-Zayer MR, Al-Sahwan NG, Al-bahrani Z, Bakry AS, Jamleh AO, Al-sayed EZ, Mandurah M, Abbassy M. Comparative SEM observation of silver-nitrate at resin-dentin interface : Nanoleakage study. Oral Health Care 2017;2:1–5.
6. Bonilla ED, Stevenson RG, Caputo AA, White SN. Microleakage resistance of minimally invasive class I flowable composite restorations. Oper Dent 2012;37:290–98.
7. Bullard RH, Leinfelder KF, Russell CM. Effect of coefficient of thermal expansion on microleakage. JADA 1988;116:871–74.
8. Camps J, Pashley D. Reliability of the dye penetration studies. J Endod 2003;29:595–99.
9. Chen X, Cuijpers VM, Fan MW, Frencken JE. Validation of micro-CT against the section method regarding the assessment of marginal leakage of sealants. Aust Dent J 2012;57:196–9.
10. Cheung GSP. Reducing marginal leakage of posterior composite resin restorations: a review of clinical techniques. J Prosth Dent 1990;63:286–90.
11. Costa JF, Siqueira WL, Logueruo AD, Resi A, Oliveira EF, Alves CMC, Bauer JRDO, Grande RHM. Characterization of aqueous silver nitrate solution for leakage tests. J Appl Oral Sci 2011;19:254–57.
12. de Goes MF, Montes MAJ. Evaluation of silver methenamine method for nanoleakage. J Dent 2004; 32:391–398.
13. de Santil Alvarenga FA, Pinelli C, de Castro Monteiro Loffredo L. Reliability of marginal microleakage assessment by visual and digital methods. Eur J Dent 2015;9:1–5.
14. Ginguanju. Nanoleakage determination techniques - A literature review. BLA 2008;8:976–79.
15. Hashimoto M, Yamaguchi S, Imazato S. Nanoleakage and durability of resin/dentin bonds. Curr Oral Health Rep 2015;2:195–201.
16. Iwani Y, Yamamoto H, Ebisu S. A new electrical method for detecting marginal leakage of in vitro resin restorations. J Dent 2000;28:241–45.

17. Iwani Y, Yamamoto H, Hayashi M, Takeshige F, Ebisu S. Validity of electrical conductance measurement in evaluating marginal leakage around resin composite restorations. Oper Dent 2002;27:606–10.
18. Labib LM, Nabih SM, Baroudi K. Nanoleakage evaluation of posterior teeth restored with low shrinkable resin composite-an invitro study. J Clin Diagn Res 2016;10: ZC102–04.
19. Mobarak E, Seyam R. Interfacial nanoleakage and bonding of self-adhesive systems cured with a modified-layering technique to dentin of weakened roots. Oper Dent 2013;38:e154–e165.
20. Moosavi H, Yazdi FM, Moghadam FV, Soltani S. Comparison of resin composite restorations microleakage: An in-vitro study Open J of Stomato 2013;3:209–14.
21. Pioch T, Staehle HJ, Duschner H, Garcia-Godoy F. Nanoleakage at the composite-dentin interface: a review Am J Dent 2001;14:252–58.
22. Sano H, Takatsu T, Ciucchi B, Homer JA. Nanoleakage: Leakage within the hybrid layer. Oper Dent 1995;20:18–21.
23. Sinescu C, Negrutiu M, Draganescu G, Todea C, Dodenciu D, Florita Z, Pop D. Microleakage in dentistry: new methods for investigating the gaps in biomaterials. Proc. of Spie: 2008;6843.
24. Swain MV, Xue J. State of the art of micro-CT applications in dental research. Int J Oral Sci 2009;1:177–88.
25. Yang H, Guo J, Guo J, Chen h, Somar M, Yue J, Huang C. Nanoleakage evaluation at adhesive-dentin interfaces by different observation methods. Dent Mater J 2015;34: 654–662.
26. Yavuz I, Aydin H. New direction for measurement of microleakage in cariology research. J Int Dent Med Res 2010;3:19–23.
27. Yavuz I, Aydin H, Ulku R, Dulgergil TC, Akdag MZ. New technique: measurement of microleakage volume in the marginal gaps of the dental restorations. Biotechnol and Biotechnol 2005;19:184–91.
28. Yoo HK, Kim SH, Kim SI, Shin YS, Shin YS and Park JW. Seven year follow-up of resin infiltration treatment on non-cavitated proximal caries. Oper Dent 2019;44:8–12.
29. Zhao XY, Li SB, Gu LJ, Li Y. Detection of marginal leakage of class V restorations in vitro by micro-computed tomography. Oper Dent 2014;39:174–80.